



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Nanyang Business School

WEEK 3

PDF Data Extraction



Considering these two invoices

ThisWebHost LTD 337 Bath Road Slough Berkshire SL1 5P United Kingdom		INVOICE # 3	
Bill To:		Date:	Apr 15, 2021
Some Company LTD - Robotic Automations 3216 Maple Court, California, MO 63101 USA		Balance Due:	\$799.80
Item	Quantity	Rate	Amount
Server Maintenance	20	\$39.99	\$799.80
		Total:	\$799.80

Invoice			
Delta Supplies Inc.			
Date: 2024-10-29			
Invoice Number: DEL-8287			
Item Description	Quantity	Unit Price	Amount
Service Charge	3	\$50.00	\$150.00
Product X	4	\$45.99	\$183.96
Consulting Fee	2	\$120.00	\$240.00
Product X	2	\$45.99	\$91.98
Total Amount:			\$665.94

I want to store their data into a spreadsheet

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But their layouts are not standardised

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Invoice

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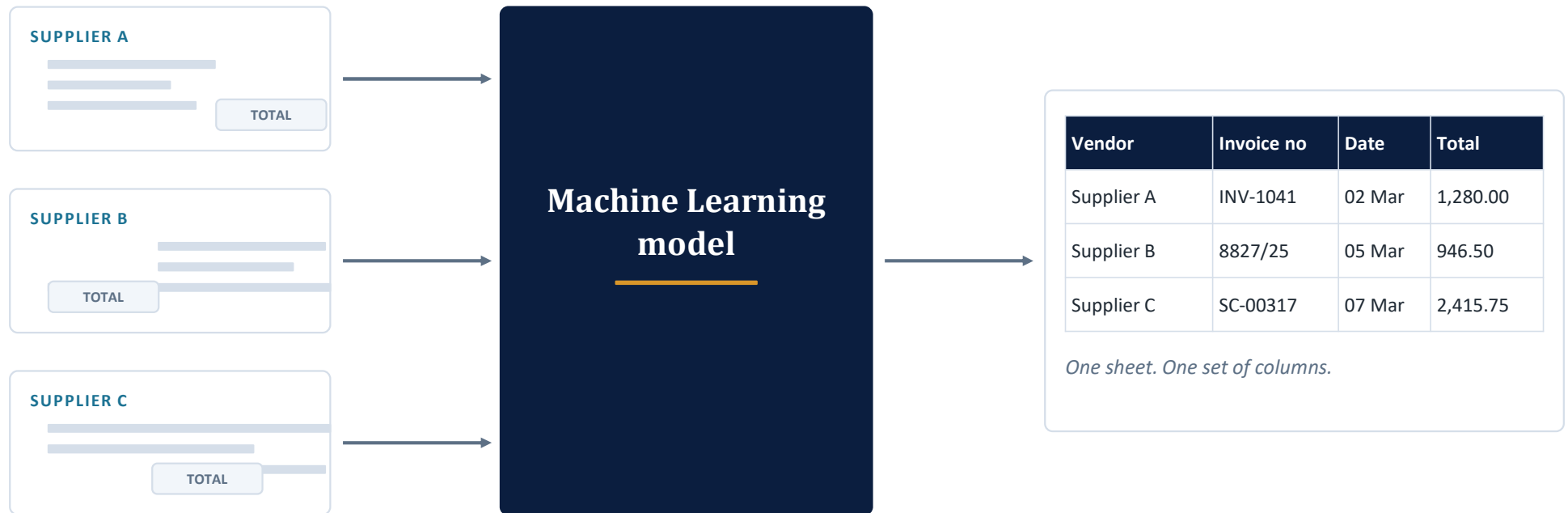
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Solution: Machine Learning

One model reads all three layouts. It looks for what a field means, not for where it sits on the page.



THE PROMISE

Three layouts in, one consistent set of fields out — with no rules written for any particular supplier.

But a model has to be trained first

A model that reads invoices did not arrive knowing what an invoice is. Somebody taught it, and that is not cheap.

What training a model takes

- 1 Collect** A few hundred real documents, covering every layout you expect to meet.
- 2 Label** Draw a box around every field, on every page, by hand. This is the slow part.
- 3 Train** Run the training job, wait for it, and pay for the compute while it runs.
- 4 Evaluate** Measure how well it reads documents it has never seen. If it is not good enough, label more and go again.

UiPath has already done it

For the document types the whole world uses — invoices, receipts, passports, bills of lading — the collecting, labelling, training and evaluating is finished. You get the trained model. You do not do the training.

Pre-trained Models in UiPath Document Understanding

Finance: AP and O2C

- Invoices, Invoices2, Receipts
- Country variants: China, India, Japan, Australia, Hebrew, Shipping
- Purchase Orders, Remittance Advices

Banking, Lending and KYC

- Bank Statements, Checks, Utility Bills
- Passports, ID Cards and Driver's Licences
- Financial Statements, FM 1003, Vehicle Titles

Tax, Payroll and HR

- 1040 and Schedules C, D, E; 1040x
- W2, W9, 4506-T, 941x, 709, 9465
- I9 employment eligibility, Payslips

Supply Chain and Shipping

- Bills of Lading, Packing Lists
- Certificate of Origin, Certificate of Incorporation
- EU Declaration of Conformity

Insurance and Healthcare

- ACORD 25, 125, 126, 131, 140
- CMS 1500 health insurance claim form
- UB04 institutional claim form

Full list: <https://docs.uipath.com/document-understanding/automation-cloud/latest/user-guide/out-of-the-box-pre-trained-ml-packages>

From what it is, to how you build it

Everything so far has been about what Document Understanding can do. The rest of the session is about the activity you actually drag onto the canvas.

COVERED SO FAR

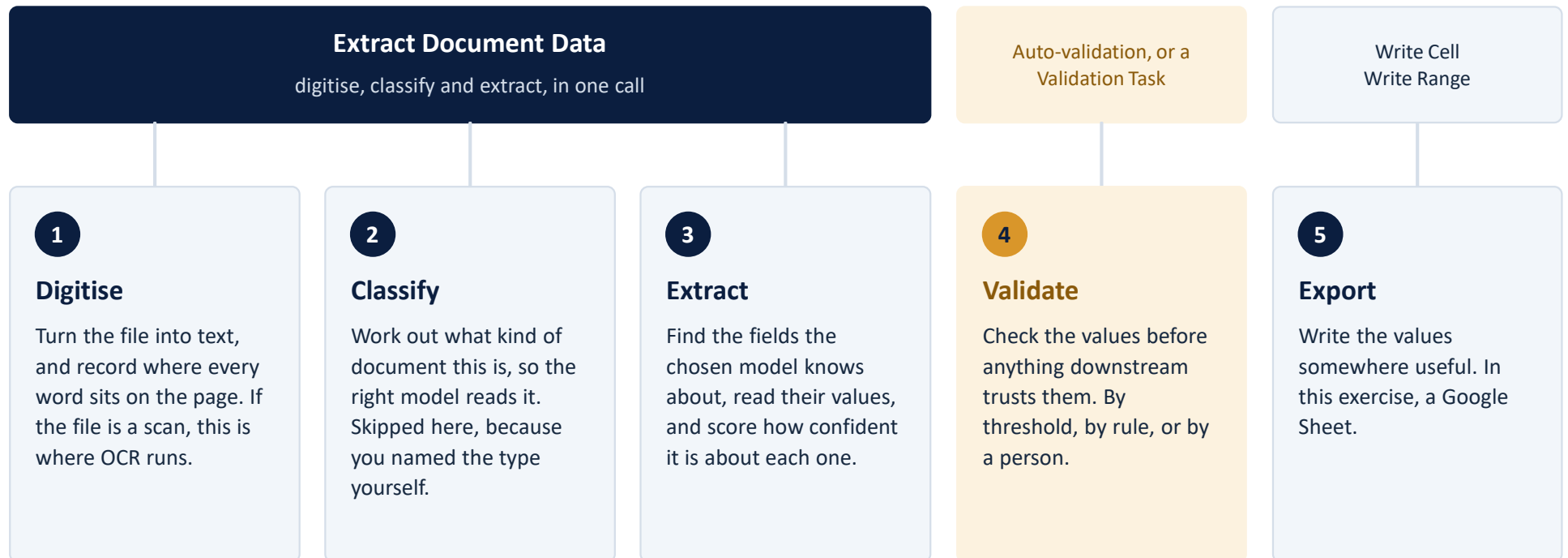
- Why two invoices with the same information defeat a rule-based approach
- What a machine learning model does instead: read for meaning, not position
- What training one costs, and who has already paid it
- Which models UiPath ships ready to call

COMING UP

- The one activity that does the extraction, and the four boxes you fill in
- What comes back, and how you reach a single value inside it
- How to choose between a fixed field list and one you write yourself
- How those values end up in a spreadsheet, one PDF at a time

One activity does most of the work. The rest of this session is knowing what to type into it.

Document data extraction steps



Extract Document Data: what you fill in

Four things in the body of the activity. Everything else has a default you can leave alone.

Extract Document Data

Input	CurrentItem	<i>the PDF handed over by the outer loop</i>
Project	Predefined	<i>the out-of-the-box family</i>
Extractor	Invoices	<i>which model inside that family</i>
Output	documentData	<i>everything the model read</i>

Input takes either a file (from For Each File or Folder) or a Document Data object (from Classify Document).

Three properties worth knowing

Generate Data Type

True gives you typed fields you can reach with a dot. False gives a plain dictionary you look up by name.

Auto-validation

Off by default. A confidence threshold of 0 means nothing is checked against anything.

Timeout (seconds)

3600, and worth leaving there. Generative calls are slow and a short timeout just fails.

What comes back: one object, typed fields

The activity returns a single Document Data object. Every field you asked for hangs off it.

EXPRESSIONS YOU CAN WRITE STRAIGHT AWAY

```
documentData.Data.VendorName.Value
```

```
documentData.Data.InvoiceNumber.Value
```

```
documentData.Data.InvoiceDate.Value
```

```
documentData.Data.Items          ' a table
```

Every field also carries .Confidence, from 0 to 1.

Value, or Confidence

Value is the text the model read. Confidence is how sure it is. Use the second to decide which documents a person needs to look at.

Where the names come from

You did not choose VendorName. The Invoices model did: 27 fields, 7 line-item columns and a currency classifier.

Set Generate Data Type to False and the dot notation disappears: you get a dictionary, and a typo becomes a run-time error instead of a red squiggle.

EXTRACT DOCUMENT DATA

INPUT

CurrentItem

PROJECT

Predefined

EXTRACTOR

Invoices

OUTPUT

documentData

One invoice, two shapes of data

Header fields have one value each. Line items are a table with as many rows as the invoice has lines.

Header fields

One value for the whole document.

- VendorName
- InvoiceNumber
- InvoiceDate
- DueDate
- TotalAmount
- Currency

```
documentData.Data.VendorName.Value  
' "Sunrise Trading Pte Ltd"
```

Line items

A table. You do not know how many rows until you look.

- Description
- Quantity
- UnitPrice
- LineAmount

```
For Each currentItems In documentData.Data.Items  
    currentItems.Description.Value  
    currentItems.Quantity.Value  
    currentItems.UnitPrice.Value  
    currentItems.LineAmount.Value
```

A loop inside a loop: one over the PDFs in the folder, one over the line items inside each PDF.

EXTRACT DOCUMENT DATA

INPUT

CurrentItem

PROJECT

Predefined

EXTRACTOR

Invoices

OUTPUT

documentData

Two ways to choose a model

Both live in the same Project dropdown. The difference is who decides what the fields are.

Predefined

the model decides the fields

- Pre-trained specialised models, ready to call
- You pick a document type: Invoices, Receipts, Purchase Orders, Bank Statements, Passports
- The field list is fixed, and published in the docs
- Nothing to write, nothing to train
- Best for documents that look broadly the same the world over

Generative Predefined

you decide the fields

- Pre-trained generative models that take instructions
- You pick by layout: Long Document Simple Layout, or the two Complex Layout extractors in Preview
- You define each field yourself, as a name and an instruction
- Best where no standard model exists: leases, service agreements, policies
- Slower, metered differently, and not available on Automation Suite

Also in that dropdown: Predefined Non-Latin Languages, and any custom project you have published in Document Understanding.

INPUT

CurrentItem

PROJECT

Predefined

EXTRACTOR

Invoices

OUTPUT

documentData

Predefined in practice: the Invoices model

You point at a document type. The schema arrives with it, whether or not you wanted every field in it.

Extract Document Data

Project

Predefined

Extractor

Invoices

No training data. No prompt.

Two dropdowns and you are extracting. That is the case for reaching here first.

FIELDS THE MODEL RETURNS (a selection)

VendorName	VendorAddress	BillingName	BillingAddress
InvoiceNumber	InvoiceDate	DueDate	PaymentTerms
PurchaseOrderNumber	NetAmount	TaxAmount	TotalAmount
Currency	VatNumber	BankAccountNumber	Iban

LINE ITEM COLUMNS (the Items table)

Description	Quantity	UnitPrice	LineAmount
-------------	----------	-----------	------------

27 fields, 7 line-item columns and a currency classifier. Read the published list before you promise a client a field.

EXTRACT DOCUMENT DATA

INPUT CurrentItem	PROJECT Predefined	EXTRACTOR Invoices	OUTPUT documentData
-----------------------------	------------------------------	------------------------------	-------------------------------

Generative Predefined in practice: you write the fields

Same activity, same output object. What changes is that you supply a name and an instruction for each thing you want.

DOCUMENT TYPE DETAILS

Field name	Instruction
Contract start date	What date does this agreement take effect?
Notice period	How many days notice must either party give to end it?
Governing law	Which country's or state's law governs this agreement?
Total value	What is the total value of the agreement, with its currency?

Limits to remember

- Field name: 30 characters
- Instruction: 1000 characters
- Response: 700 words per document
- Past that, split the document, extract, then merge

When to reach for it

No pre-trained model covers the document, the layout varies, and the fields are things a person would have to read for.

The output object does not change. A careless field name here becomes a careless name in every expression downstream.

EXTRACT DOCUMENT DATA

INPUT

CurrentItem

PROJECT

Predefined

EXTRACTOR

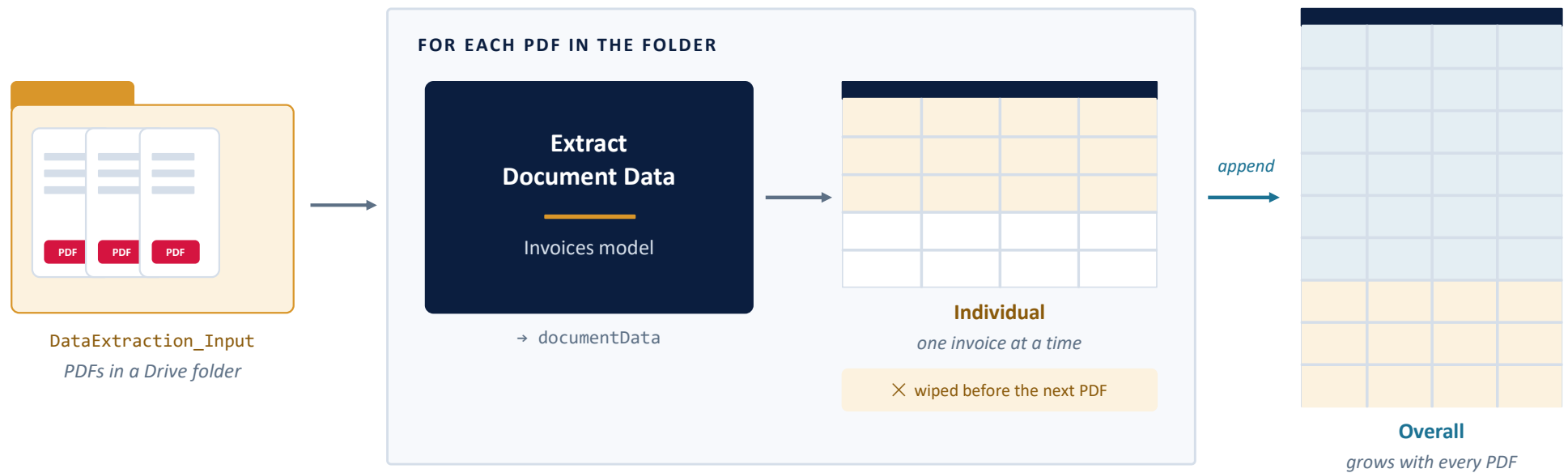
Invoices

OUTPUT

documentData

From a folder of PDFs to one growing sheet

The whole workflow in one picture. Everything inside the box happens once per PDF.



One invoice, four moves on the sheets

The same two sheets, photographed after each activity. Watch the Individual sheet fill, empty into the master, and clear.



Nothing errors if you leave step 5 out. The Overall sheet just quietly doubles up, one invoice at a time.

The cell map on the Individual sheet

Header fields sit on row 1. Line items run down columns D to G, one row per item.

Individual							
	A	B	C	D	E	F	G
1	VendorName	InvoiceNumber	InvoiceDate	Description	Quantity	UnitPrice	LineAmount
2				Description	Quantity	UnitPrice	LineAmount
3				Description	Quantity	UnitPrice	LineAmount

Grey is written once per invoice. **Amber** is written once per line item — the same block you watched move across on the last slide.

Three writes for the header, four per line item. An invoice with three lines takes fifteen Write Cell calls.

HOW THE CELL IS ADDRESSED

Cell "A1"
Value documentData.Data
.VendorName.Value

Cell "D" & CurrentRowNumber
Value currentItems.Description.Value

Why index + 1

For Each hands you a zero-based index. Spreadsheet rows start at 1. CurrentRowNumber = index + 1 is the whole reason that Assign is there.

PITFALL

Vendor, invoice number and date are written once, on row 1 of each block. After the append, the Overall sheet carries those three values only on the first row of each invoice. Either fill down afterwards, or write them inside the item loop so every row stands on its own.

Straight through processing, and when a person steps in

Nothing you have built so far decides which documents a person looks at. That is a design decision, and it is yours.

STP

Straight through processing

Every field the workflow depends on clears its confidence threshold and the arithmetic checks pass. The document is read, written to the sheet and closed with nobody looking at it.

The share of documents that get this far is your STP rate.

HITL

Human in the loop

One field falls short, or a rule fails. The document is held, and a person is shown what the model read next to the page it read it from. They confirm or correct, and the run carries on.

In UiPath this is a validation task waiting in Action Center.

WHAT YOU HAVE TO DECIDE

1

Where the threshold sits

Too high and everything queues.
Too low and a wrong number reaches the ledger. There is no correct value in the docs.

2

Risk differs by field

Set the bar field by field. A wrong bank account number costs far more than a wrong payment term.

3

Confident is not correct

A model can be sure and wrong.
Check that line items sum to the net, and net plus tax to the total.

4

Someone clears the queue

Volume times exception rate is a daily workload. A queue nobody is staffed for just moves the delay.

Every correction a person makes is labelled data. Feed it back and the exception rate falls; ignore it and next year looks exactly like this one.